

Matter waves:-

De Broglie put forward the view that matter light radiation should exhibit dual nature i.e. the matter which is ordinarily considered as made up of discrete particles like proton, electrons etc. might exhibit wave like properties under applied condition.

This wave which is possessed by matter is called matter wave. It is also known as de-broglie wave. In order to put forward his view of matter wave. He took into account the following facts:-

i) It is found that radⁿ possess a dual nature and hence matter also inherit the same dual nature because of the existence of symmetry in nature.

ii) The basic law of governing the motion of material object which is principle of least action & the wave motion which is fermat's principle of least time as similar forms. This leads one to believe that the two forms matter & radiation have some property in common & a relation can be sorted. De Broglie proposed that energy of the particle motion must be carried by some sort of wave associated with particles just like wave associated with wave photon carry energy.

iii) It is obvious from Bohr's assumption that the stable non radiating orbit (stationary) of the e^- in the atoms are governed by integer rule i.e. radii of this orbits are proportional to the sq. of integral numbers. De Broglie gave the satisfactory explanation of this assumption. The only phenomena involving integers where those of interference & nodes of vibration of stretched string. Both imply wave motion.

Properties of matter waves:-

1) Matter waves are not physical phenomena but a wave probability. The wavelength associated with matter waves decreases with increase in the mass of the particle.

$$\text{i.e. } \lambda = \frac{h}{\sqrt{2mcv}}$$

2) Matter waves are new kind of waves and are not electromagnetic in nature.

3) Matter waves are mainly used as a wave to guide matter particles and hence are known as guide waves.

4) The velocity of matter waves depends on the velocity of material particle.

5) The phase velocity of a matter wave is inversely prop. to its wavelength.

6) Matter waves are not constant as electromagnetic wave.

Physical significance of wave function:-

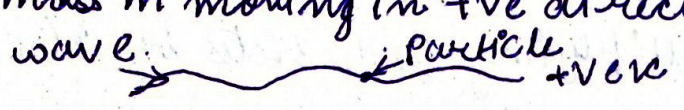
1) The qty. whose variation makeup matter waves is called the wave f^n . The value of the wave f^n associated with a moving body at the particular point (x, y, z) in space at time t is related to the likely hood of finding the body there at the time. The wave $f^n(\psi)$ itself however has no direct physical significance. However following are the physical significance of ψ assigned to it by different physicist.

2) A/c to millikan it is an auxiliary qty introduced to facilitate computation related to experimental results but this physical significance is not in the speed

of theoretical physics to introduce an isolated mathematical f^n .

3/ Schrodinger gave the interpretation of Ψ in terms of charge density and said that if Ψ is the amplitude of matter waves in any point of space then the particle density of that point may be taken as proportional to $|\Psi|^2$. Thus the sq. of absolute of Ψ is a measure of the particle density. The charge density obtained by multiplying the particle density by charge of the particle. Since charge is constant. Therefore $|\Psi|^2$ is a measure of charge density.

Schrodinger Time - dependent eqⁿ:-

Consider a particle of mass m moving in +ve direction of x -axis with velocity v . 

Acc. to de broglie hypothesis, every moving particle has a wave associated with it which is called a matter wave whose wavelength is given by $\lambda = \frac{h}{mv} = \frac{h}{p_x}$

$$\Rightarrow p_x = \frac{h}{\lambda} \Rightarrow \frac{h}{2\pi} \times \frac{2\pi}{\lambda} = \hbar k$$

$$p_x = \hbar k \Rightarrow \boxed{k = \frac{p_x}{\hbar}} \quad - (1)$$

Acc. to quantum physics, a wave consists of tiny packets of energy called photons. The energy of each photon is given by

$$E = h \nu \Rightarrow \frac{h}{2\pi} \times 2\pi \nu \Rightarrow \hbar \omega$$

$$\Rightarrow E = \hbar \omega \Rightarrow \boxed{\omega = \frac{E}{\hbar}} \quad - (2)$$

The wave associated with moving particle can be represented by a wave function $\Psi(x, t)$ as

$$\Psi = \boxed{\Psi(x, t) = A e^{-i(\omega t - kx)}} \quad \text{--- (3)}$$

$A \rightarrow \text{Amplitude}$
 $\omega \rightarrow \text{Ang. freq.}$
 $k \rightarrow \text{propagation const}$

Using ① & ② in ③

$$\begin{aligned} \Psi(x, t) &= A e^{-i\left(\frac{E}{\hbar} t - \frac{p_x}{\hbar} \cdot x\right)} \\ &= A e^{i/\hbar (p_x x - Et)} \quad \text{--- (4)} \end{aligned}$$

Now total energy E of the particle

$$E = KE + PE$$

$$= \frac{p_x^2}{2m} + U$$

$$KE = \frac{1}{2} m v^2$$

$$= \frac{p_x^2}{2m}$$

Multiplying both sides by $\Psi(x, t)$

$$E \Psi(x, t) = \frac{p_x^2}{2m} \Psi(x, t) + U \Psi(x, t) \quad \text{--- (5)}$$

Diff w.r.t t

$$\frac{\partial \Psi(x, t)}{\partial t} = A e^{i/\hbar (p_x x - Et)} \cdot \frac{-i(E)}{\hbar}$$

$$\frac{\partial \Psi}{\partial t} = \Psi \times \frac{i(-E)}{\hbar} \times \frac{i}{i}$$

$$i\hbar \frac{\partial \Psi}{\partial t} = \Psi E \quad \text{--- (6)}$$

Diff ④ w.r.t x

$$\frac{\partial \Psi}{\partial x} = A e^{i/\hbar (p_x x - Et)} \cdot \frac{i p_x}{\hbar}$$

$$\frac{\delta \psi}{\delta x} = \psi \cdot \frac{i p x}{\hbar} \quad \text{--- (7)}$$

Differentiating (7) w.r.t x

$$\frac{\delta^2 \psi}{\delta x^2} = \frac{\delta \psi}{\delta x} \times \frac{i p x}{\hbar} \quad \text{--- (8)}$$

Using (7) in (8)

$$\frac{\delta^2 \psi}{\delta x^2} = \psi \left(\frac{i p x}{\hbar} \right)^2$$

$$\frac{\delta^2 \psi}{\delta x^2} = - \frac{\psi (p x^2)}{\hbar^2}$$

$$\psi p x^2 = - \hbar^2 \frac{\delta^2 \psi}{\delta x^2} \quad \text{--- (9)}$$

Using (6) & (9) in (5)

$$\psi E = \frac{p x^2}{2m} \psi + U \psi$$

$$\boxed{i \hbar \frac{\delta \psi}{\delta t} = - \frac{\hbar^2}{2m} \frac{\delta^2 \psi}{\delta x^2} + U \psi}$$

This is Schrodinger time-dependent wave eqⁿ.

$$\text{In 3-D } i \hbar \frac{\delta \psi}{\delta t} = - \frac{\hbar^2}{2m} \left[\frac{\delta^2 \psi}{\delta x^2} + \frac{\delta^2 \psi}{\delta y^2} + \frac{\delta^2 \psi}{\delta z^2} \right] + U \psi$$

$$\boxed{i \hbar \frac{\delta \psi}{\delta t} = - \frac{\hbar^2}{2m} \nabla^2 \psi + U \psi}$$

Time-independent Schrodinger wave eqⁿ (Steady state eqⁿ)

Consider a particle of mass 'm' moving with velocity 'v' along +ve x dirⁿ. Let P & E be the momentum & energy of the particle resp.

According to de-broglie hypothesis every moving particle is associated with a wave called matter wave. The wave function describing the behaviour of wave associated with moving particle can be expressed as:-

$$\Psi(x, t) = \Psi = A e^{i/\hbar (p_x x - Et)} \quad \text{--- (1)}$$

We know that Schrodinger time-dependent

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + U(\Psi) \quad \text{--- (2)}$$

Diff. eqⁿ (1) w.r.t 't'

$$\frac{\partial \Psi}{\partial t} = A e^{i/\hbar (p_x x - Et)} \times \frac{i(-E)}{\hbar}$$

$$\frac{\partial \Psi}{\partial t} = -\Psi \times \frac{iE}{\hbar}$$

$$\frac{\partial \Psi}{\partial t} = \frac{\Psi E}{\hbar i} \Rightarrow \boxed{i\hbar \frac{\partial \Psi}{\partial t} = \Psi E} \quad \text{--- (3)}$$

Let potential energy U of the particle depends only on position x & does not depend on time 't'. In this case wave function $\Psi(x, t)$ can be written as:-

$$\Psi(x, t) = \phi(x) \cdot v(t) \leftarrow \text{Method of separation of variable}$$

$$\boxed{\Psi = \phi u} \quad \text{--- (4)}$$

Differentiate ④ wrt x

$$\frac{\delta \psi}{\delta u} = u \frac{\delta \phi}{\delta x}$$

$$\frac{\delta^2 \psi}{\delta u^2} = u \frac{\delta^2 \phi}{\delta u^2} \quad \text{--- (5)}$$

using ③, ④, ⑤ in ②

$$E \psi = -\frac{\hbar^2}{2m} \left(u \cdot \frac{\delta^2 \phi}{\delta u^2} \right) + u (\phi \phi)$$

$$E \phi u = -\frac{\hbar^2}{2m} u \cdot \frac{\delta^2 \phi}{\delta u^2} + u \phi u$$

$$E \phi = -\frac{\hbar^2}{2m} \frac{\delta^2 \phi}{\delta u^2} + u \phi$$

$$\boxed{\frac{\hbar^2}{2m} \frac{\delta^2 \phi}{\delta u^2} + (E - U) \phi = 0}$$

This is Schrodinger time independent wave eqⁿ.

$$\frac{\hbar^2}{2m} \left[\frac{\delta^2 \phi}{\delta u^2} + \frac{\delta^2 \phi}{\delta y^2} + \frac{\delta^2 \phi}{\delta z^2} \right] + (E - U) \phi = 0$$

$$\boxed{\frac{\hbar^2}{2m} \nabla^2 + (E - U) \phi = 0}$$

Operator:-

An operator in general is an expression that acts on a function to produce a same or diff f^n . If the same f^n is reproduced the eqⁿ is called an eigen value eqⁿ. An operator tells us what operation to carry out on the quantity that follows it. It is customary to denote operators by caret.

Hence, required eigen value eqⁿ can be written as

$$\hat{A} \psi = \lambda \psi \rightarrow \text{eigen value.} \quad \left| \begin{array}{l} \hat{A} \psi \rightarrow \text{operator acting on } \psi \end{array} \right.$$

↪ eigen operator

Operator corresponding to energy:-

The simplest type of a wave is plane monochromatic wave for described by wave function.

$$\psi(x, t) = A e^{i(kx - \omega t)}$$

$$\frac{\partial \psi(x, t)}{\partial t} = -i \omega \psi(x, t)$$

$$\frac{\partial \psi(x, t)}{\partial t} = -i \omega \psi$$

$$= -i \times \frac{E}{\hbar} \times \frac{\hbar}{i} \times \psi$$

$$= \frac{E \psi}{i \hbar}$$

$$\text{or } i \hbar \frac{\partial \psi(x, t)}{\partial t} = E \psi$$

$$\text{Energy operator} = i \hbar \frac{\partial \psi}{\partial t}$$

$$\boxed{\hat{E} = i \hbar \frac{\partial \psi}{\partial t}}$$

Expectation Value.

Expectation value of a quantity is the average value of the measurement of the quantity performed in very large no. of independent identical system represented by wave function ψ or equivalently it is the average of the large no. of measurement for the same system.

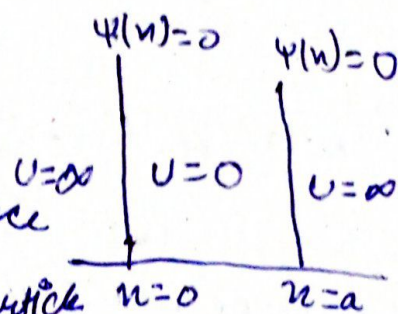
Ex - In three dimension an example would be a set of measurement of posn of e^- in hydrogen atom with all the atoms in their lowest energy state.

Motion of a free particle (1-D) or square well or infinite potential well in 1D.

In quantum mechanics a box means

a region of space in which particle experience no force i.e. the potential energy of the particle

is zero in this region and outside there potential energy is infinite. 1D box means that the particle is allowed to move only along a straight line say along x -axis.



Consider a particle of mass ' m ' confined to move in 1-D box of length a along x -axis under following conditions:-

- i) The wall of the box are rigid and perfectly elastic so that the particle rebounded with same $K.E$ after making elastic collision with the wall and the total energy E of the particle remains constant.
- ii) The wall of the box are non-penetrable so that there is no probability of finding the particle outside of the box.

Potential energy of $U(x)$ for 1-D box

$$U(x) = \begin{cases} 0 & ; 0 < x < a \\ \infty & ; x < 0 \text{ \& } x > a \end{cases} \quad \text{--- (1)}$$

Schrodinger time independent of wave eqⁿ

$$\boxed{\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - U) \psi = 0}$$

for $0 < x < a$ we have $U(x) = 0$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E \psi = 0$$

$$\left\{ \frac{2m}{\hbar^2} E = k^2 \right.$$

$$\boxed{\frac{d^2\psi}{dx^2} + k^2 \psi = 0} \quad \text{--- (2)}$$

$$K = \frac{\sqrt{2mE}}{\hbar} \quad - (3)$$

General solution of eqⁿ (2) will be

$$\psi(x) = A \sin Kx + B \cos Kx \quad - (4)$$

where A & B are arbitrary constant

$$\text{At } x=0, \psi(x) = 0$$

$$0 = 0 + B$$

$$B = 0 \quad - (5)$$

Putting (5) in (4)

$$\psi(x) = A \sin Kx \quad - (6)$$

$$\text{At } x=a, \psi(x) = 0$$

$$\psi(a) = A \sin Ka$$

$$A \sin Ka = 0 \quad A \neq 0$$

$$Ka = n\pi$$

$$K = \frac{n\pi}{a} \quad - (7)$$

From (3) & (7)

$$\frac{n\pi}{a} = \frac{\sqrt{2mE}}{\hbar}$$

$$\frac{n^2 \pi^2}{a^2} = \frac{2mE}{\hbar^2}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \quad - (8)$$

$$n=1$$

$$E = \frac{\pi^2 \hbar^2}{2ma^2} \quad (\text{Zero point energy})$$

$$n=2$$

$$E = \frac{4\pi^2 \hbar^2}{2ma^2}$$

Here the particle is quantised.

for eigen function for particle in 1-D box,

$$\psi(x) = A \sin kx \quad (\text{from } (6))$$

$$= \int_0^a \psi^*(x) \psi(x) dx = 1$$

$$= \int_0^a A \sin kx A \sin kx dx = 1$$

$$= A^2 \int_0^a \sin^2 kx dx = 1, \quad k = \frac{n\pi}{a}$$

$$= A^2 \int_0^a \sin^2 \frac{n\pi}{a} x dx = 1$$

$$= A^2 \int_0^a \frac{1 - \cos\left(\frac{2n\pi x}{a}\right)}{2} dx = 1$$

$$= \frac{A^2}{2} \int_0^a 1 - \cos\left(\frac{2n\pi x}{a}\right) dx = 1$$

$$= \frac{A^2}{2} [a - 0] = 1 \Rightarrow \boxed{A = \sqrt{\frac{2}{a}}} \quad - (9)$$

using (9) in $\psi(x)$

eigen function $\boxed{\psi(x) = \sqrt{\frac{2}{a}} \sin kx} \quad - (10)$

Nodes & Position of nodes:- Within the box there are certain points at which $\psi(x) = 0$. Such points are nodes.

$$\psi(x) = 0$$

$$\sqrt{\frac{2}{a}} \sin kx = 0$$

$$\sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x = 0$$

$$\sin \frac{n\pi}{a} x = \sin p\pi \Rightarrow \boxed{x = \frac{pa}{n}}$$

$p = 0, 1, 2, \dots$

$$p=0, 1, 2, \dots$$

$$n=1 \quad u=0, a$$

$$n=2, \quad u=0, a/2, a$$

$$n=3, \quad u=0, \frac{a}{3}, \frac{2a}{3}, a, \dots$$

Antinode with probability of finding e^- is

$$\psi(u) = n a x$$

$$\psi(u) = 1$$

$$\sin kx = \sin (2p+1) \frac{\pi}{2}$$

$$\frac{n\pi x}{a} = (2p+1) \frac{\pi}{2}$$

$$x = \frac{(2p+1)a}{2n}$$

$$p=0$$

$$p=1$$

$$p=2$$

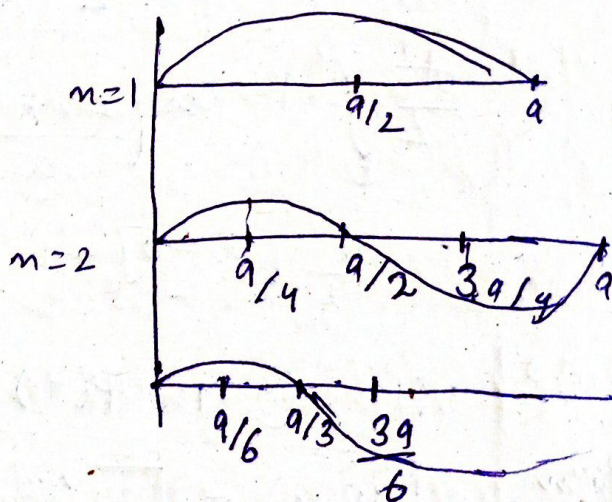
$$n=1 \quad \frac{a}{2}$$

$$n=2 \quad \frac{a}{4}$$

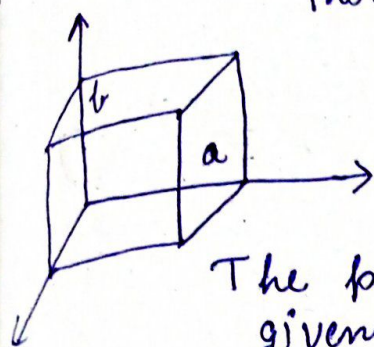
$$n=3 \quad \frac{a}{6}$$

$$\frac{3a}{4}$$

$$\frac{3a}{6}$$



Particle in 3-D box (e^- in infinite deep potential well)
 inside box potential energy is zero.



Consider a particle of mass ' m ' confined in rectangular box having sides a , b and c along x , y and z direction respectively.

The potential energy function $V(x)$ for a 3-D box is given

$$V(x) = \left\{ \begin{array}{l} 0; \quad 0 < x < a \\ 0; \quad 0 < y < b \\ 0; \quad 0 < z < c \end{array} \right\} \text{ inside the box}$$

$\infty \rightarrow$ outside the box

The time independent Schrodinger wave eqⁿ is 3D given as

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0 \quad \text{--- (1)}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi + \frac{2m}{\hbar^2} E \psi = 0 \quad \text{--- (2)}$$

($V=0$) $\rightarrow$ inside the box

$$\text{Let } \psi(x, y, z) = \psi_1(x) \psi_2(y) \psi_3(z) \quad \text{--- (3)}$$

Using eqⁿ (3) in eqⁿ (2) we get

$$\frac{\partial^2}{\partial x^2} (\psi_1 \psi_2 \psi_3) + \frac{\partial^2}{\partial y^2} (\psi_1 \psi_2 \psi_3) + \frac{\partial^2}{\partial z^2} (\psi_1 \psi_2 \psi_3) + \frac{2m}{\hbar^2} E \psi_1 \psi_2 \psi_3 = 0$$

$$\psi_2 \psi_3 \frac{\partial^2 \psi_1}{\partial x^2} + \psi_1 \psi_3 \frac{\partial^2 \psi_2}{\partial y^2} + \psi_1 \psi_2 \frac{\partial^2 \psi_3}{\partial z^2} + \frac{2mE}{\hbar^2} = 0 \quad \text{--- (4)}$$

$$\text{Here, } E = E_x + E_y + E_z \quad \text{--- (5)}$$

Putting eqⁿ (5) in (4)

$$\frac{1}{\psi_1} \frac{\partial^2 \psi_1}{\partial x^2} + \frac{1}{\psi_2} \frac{\partial^2 \psi_2}{\partial y^2} + \frac{1}{\psi_3} \frac{\partial^2 \psi_3}{\partial z^2} + \frac{2m}{\hbar^2} (E_x + E_y + E_z) = 0$$

$$\Rightarrow \frac{1}{\psi_1} \frac{\partial^2 \psi_1}{\partial x^2} + \frac{1}{\psi_2} \frac{\partial^2 \psi_2}{\partial y^2} + \frac{1}{\psi_3} \frac{\partial^2 \psi_3}{\partial z^2} + \frac{2m}{\hbar^2} E_x + \frac{2m}{\hbar^2} E_y + \frac{2m}{\hbar^2} E_z = 0$$

$$\left(\frac{1}{\psi_1} \frac{\partial^2 \psi_1}{\partial x^2} + \frac{2m}{\hbar^2} E_x \right) + \left(\frac{1}{\psi_2} \frac{\partial^2 \psi_2}{\partial y^2} + \frac{2m}{\hbar^2} E_y \right) + \left(\frac{1}{\psi_3} \frac{\partial^2 \psi_3}{\partial z^2} + \frac{2m}{\hbar^2} E_z \right) = 0$$

$$\frac{1}{\psi_1} \frac{\partial^2 \psi_1}{\partial x^2} + \frac{2m}{\hbar^2} E_x = 0$$

multiply by ψ_1 in eqⁿ

$$\frac{\partial^2 \psi_1}{\partial x^2} + \frac{2m E_x \psi_1}{\hbar^2} = 0 \quad \text{--- (6)}$$

$$\frac{1}{\psi_2} \frac{\partial^2 \psi_2}{\partial y^2} + \frac{2m E_y}{\hbar^2} = 0$$

multiply by ψ_2

$$\frac{\partial^2 \psi_2}{\partial y^2} + \frac{2m E_y \psi_2}{\hbar^2} = 0 \quad \text{--- (7)}$$

$$\frac{1}{\psi_3} \frac{\partial^2 \psi_3}{\partial z^2} + \frac{2m}{\hbar^2} E_z = 0$$

Multiply by ψ_3 in eqⁿ

$$\frac{\partial^2 \psi_3}{\partial z^2} + \frac{2m E_z \psi_3}{\hbar^2} = 0 \quad \text{--- (8)}$$

$$\text{Put } \frac{2m E_x}{\hbar^2} = k^2 x \quad \text{--- (9)}$$

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} + k^2 x \psi = 0} \quad \text{--- (10)}$$

Eqⁿ (10) is 2nd order partial differential eqⁿ whose general solⁿ is given by $\psi(x) = A \sin k_n x + B \cos k_n x$ --- (11)

where A & B are arbitrary constant

$$\text{At } x=0 \quad \psi(x)=0$$

$$0 = 0 + B$$

$$B = 0 \quad \text{--- (12)}$$

using Eqⁿ (12) in Eqⁿ (11) we get.

$$\psi_1(x) = A \sin k_n x \quad \text{--- (13)}$$

At $x=0$ $\psi_1(u)=0$
 $0 = A \sin k_n a$ where $A \neq 0$
 $\sin k_n a = 0$
 $k_n a = n\pi \Rightarrow \boxed{k_n = \frac{n\pi}{a}} \quad - (14)$

On eq. both sides we get.

$$k_n^2 = \frac{n^2 \pi^2}{a^2} \quad - (15)$$

comparing eqⁿ (9) & (15)

$$\frac{2m E_n}{\hbar^2} = \frac{n^2 \pi^2}{a^2}$$

$$E_n = \frac{n^2 \pi^2}{a^2} \times \frac{\hbar^2}{2m} \quad - (16)$$

from eqⁿ (13)

$$\psi_1(u) = A \sin k_n x$$

According to normalization condition

$$\int_a^b \psi_1^*(u) \psi(u) du = 1$$

$$\int_a^b A \sin k_n x \cdot A \sin k_n x dx = 1$$

$$A^2 \int_a^b \sin^2 k_n x dx = 1$$

$$A^2 \int_0^a \sin^2 \frac{n\pi x}{a} dx = 1$$

using Trigonometry method

$$\frac{A^2}{2} [a - 0] = 1$$

$$\boxed{A = \sqrt{\frac{2}{a}}}$$

$$\text{so, } \psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi x}{a}\right) \quad \text{--- (17)}$$

wave function for the particle moving along the x-direction,
and we get corresponding eigen values.

$$E_x = \frac{n_x^2 \pi^2 \hbar^2}{2ma^2} \quad \text{--- from eq}^n \text{ (6)}$$

Similarly on solving eq (7) we get

$$\psi_2(y) = \sqrt{\frac{2}{b}} \sin\left(\frac{n_y \pi y}{b}\right) \quad \text{--- (18)}$$

$$E_y = \frac{n_y^2 \pi^2 \hbar^2}{2mb^2} \quad \text{--- (19)}$$

Similarly on solving eq (8) we get

$$\psi_3(z) = \sqrt{\frac{2}{c}} \sin\left(\frac{n_z \pi z}{c}\right) \quad \text{--- (20)}$$

$$E_z = \frac{n_z^2 \pi^2 \hbar^2}{2mc^2} \quad \text{--- (21)}$$

Here we have to multiply these wave function to get final wave function for a particle in 3-D box.

$$\text{As } \Psi = \psi_1(x) \psi_2(y) \psi_3(z)$$

$$\Psi = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right) \quad \text{--- (22)}$$

$$\text{and } E = E_x + E_y + E_z.$$

$$E = \frac{\pi^2 \hbar^2}{2m} \left[\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right] \quad \text{--- (23)}$$

eqⁿ (22) and (23) represents eigen function and energy
eigen values for a particle in a 3-D rectangular box

Finite Potential Well:-

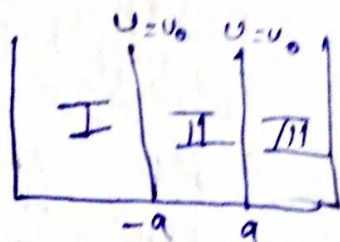
Two boundaries where a finite force prevents a particle moving freely from escaping is called the finite square well.

The Hamiltonian is given by

$$H = T + U$$

for particle;

$$U_x = \begin{cases} 0; & -a \leq x \leq a \\ U_0; & x < -a \text{ or } x > a \end{cases}$$



For region I & III

- Region - I ($x < -a$) & Region III ($x > a$)

In this region potential energy is non-zero ($U = U_0$)

so, Schrodinger wave eqⁿ becomes

$$\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + (E - U_0) \psi = 0$$

$$\text{or } -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = (E - U_0) \psi$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{2m}{\hbar^2} (E - U_0) \psi$$

where,

$$k^2 = -\frac{2m}{\hbar^2} (E - U_0)$$

$$\boxed{k^2 \psi = \frac{\partial^2 \psi}{\partial x^2}}$$

The general solutions to this differential equation in region ① & ③ are.

$$\psi_1(x) = A e^{-kx} + B e^{kx}$$

The 1st term in ψ_1 becomes non-physical for $x \rightarrow -\infty$.

So the solⁿ in this region is $\psi_1(x) = B e^{kx}$ — ①

$$\psi_2(x) = C e^{-kx} + D e^{kx}$$

Similarly the 2nd term becomes non physical for $x \rightarrow \infty$

$$\psi_3(x) = C e^{-kx} \text{ — ②}$$

for region ②

$$-a < x < a$$

$$U = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} E \psi = 0$$

$$k^2 = \frac{2mE}{\hbar^2}$$

$$\frac{\partial^2 \psi}{\partial x^2} = -k^2 \psi$$

General solⁿ to this differential is

$$\psi_2(x) = F \sin kx + G \cos kx \quad \text{--- (3)}$$

At boundary condⁿ

$$x = -a$$

$$\psi_1(-a) = \psi_2(-a) \cdot \frac{\partial \psi_1(-a)}{\partial x} = \frac{\partial \psi_2(-a)}{\partial x}$$

• Similarly at $x = a$ the boundary condⁿ

$$\psi_2(a) = \psi_3(a) \cdot \frac{\partial \psi_2(a)}{\partial x} = \frac{\partial \psi_3(a)}{\partial x}$$

• These eqⁿ have two types of solutions.

Symmetric & anti-symmetric.

- The anti-symmetric solⁿ for boundary $x = -a$ using eqⁿ ① & eqⁿ ③ we get,

$$B e^{-ka} = -F \sin ka; \quad B k e^{-ka} = -F k \cos ka$$

Dividing the both eqⁿ $k = -\cot ka$

- The symmetric solⁿ for boundary $x = a$ using eqⁿ ② & eqⁿ ③

$$C e^{-ka} = G \cos ka;$$

$$-C k e^{-ka} = -G k \sin ka$$

$$K = l \tan la$$

Dividing both eq^{ns} in the form of energy.

$$\text{constant } K \text{ is given by } K^2 = -\frac{2m}{\hbar^2} (E - U_0)$$

On Rearranging the eqⁿ we get energy

$$E = U_0 - \frac{\hbar^2 K^2}{2m}$$

Since $K = l \tan la$
Symmetric.

$$E = U_0 - \frac{A^2 \hbar^2 \tan^2 la}{2m}$$

If we put, $l^2 = \frac{2mE}{\hbar^2}$, we get another two values of energy.

$K = -l \cot la$
Antisymmetric

$$E = U_0 + \frac{A^2 \hbar^2 \cot^2 la}{2m}$$